Chapter 8 · Data types and preprocessing

Two students have marks 40 and 80. If we measure achievement on a different scale, the underlying ordering can remain the same. But assigning labels such as Low and High loses numerical detail. These are different transformations, and the exam may ask exactly what each preserves.

Vocabulary. Nominal categories have no inherent order. Ordinal categories have an order but not necessarily equal gaps. Interval scales support meaningful differences; their zero is arbitrary. Ratio scales also have a meaningful zero. Discrete/continuous describes a different distinction. Numeric category codes do not automatically permit meaningful averaging.

R37 Numeric normalization

Identify the specified rule and statistics. Min-max mapping to [a,b] is a+(x-min)(b-a)/(max-min). Z-score is (x-mean)/standard deviation. Decimal scaling divides by 10^j under the specified size convention. Substitute with units consistently and round only at the end. Constant-valued data requires an explicit handling rule for a zero denominator. Population and sample standard deviations are different if the question requires computing one.

Worked example W37. A department code labels a category, so it is nominal even if stored as an integer. A satisfaction rating Low/Medium/High is ordinal: it has order but unspecified gaps. Celsius temperature is interval: differences are meaningful but zero is not absence of temperature. Mass in kg is ratio: its zero and ratios have physical meaning. Discrete versus continuous is a separate distinction. For numeric x=60, minimum 20, maximum 80, mean 50 and standard deviation 10, min-max to [0,1] gives (60−20)/(80−20)=2/3. Z-score gives (60−50)/10=1, meaning one standard deviation above the mean. To [-1,1], the result is −1+2×(2/3)=1/3. Decimal scaling, with the smallest j making max|x|/10^j<1 and maximum absolute value 80, uses j=2 and gives 0.60. Constant data make min-max's denominator zero; the question must specify a handling convention. Numeric normalization changes a scale; relational normalization changes a schema under dependencies.

Check before moving on. Name the scale, transformation and statistics. Do not divide by zero or average arbitrary category codes.

R38 Discretization and bin smoothing

Sort the data. Equal-width bins divide the numeric range evenly; equal-frequency bins aim for similar counts. State how boundary values and ties are assigned. A question may supply that convention. For smoothing, replace values by the bin mean, median or nearest boundary as specified; preserve repeated observations in those calculations. Discretization replaces values with bin labels. Smoothing and labelling are different operations.

Worked example W38. Sort 0,1,2,3,8,9. Equal-width bins [0,3),[3,6),[6,9] contain {0,1,2},{3},{8,9}. Their means are 1,3,8.5, so smoothing gives 1,1,1,3,8.5,8.5. Equal-frequency bins with two observations each are {0,1},{2,3},{8,9}; their means are 0.5,2.5,8.5. Smoothing gives 0.5,0.5,2.5,2.5,8.5,8.5. Replacing observations with labels Bin1/Bin2/Bin3 is discretization, not mean smoothing. Boundary and tie rules must be explicit: a value 3 goes into the second stated equal-width interval. For boundary smoothing using a bin's observed minimum and maximum, replace each member with its nearer boundary and follow the stated tie rule.

Check before moving on. State intervals and ties, preserve repeated observations, and distinguish numeric smoothing from bin labels.

R39 Sampling

Identify with/without replacement and whether selection is simple, stratified, systematic or cluster-based. Determine the sampling unit before counting probabilities. For strata, use their actual sizes and sampling allocations; equal numbers from unequal strata do not make an unweighted sample mean representative. Do not import general Markov-chain sampling machinery merely because the DA preprocessing syllabus includes sampling.

Worked example W39. A population has 90 members in stratum A and 10 in B. We sample five from each stratum and observe sample means 10 and 30. The population-mean estimate is $0.9\times10+0.1\times30=12$. The raw sample mean is $(5\times10+5\times30)/10=20$; equal sample allocation overrepresents B unless weights are applied. A simple random sample chooses individual units uniformly. A stratified design samples within each stratum. A cluster design might choose villages and include every household in the chosen villages; the initial sampling unit is a village. A systematic design takes every kth unit after a random start; periodic ordering can make a realized sample unrepresentative. For two ordered draws from five IDs, sampling without replacement gives $5\times4=20$ outcomes; with replacement gives $5\times5=25$. Do not confuse an ordered sampling sequence with an unordered subset.

Check before moving on. Identify the sampling unit, replacement rule, stratum sizes and weights before calculating.

R40 Compression

Distinguish exact reconstruction from an approximation: lossless versus lossy. For a stated encoding, count actual encoded bits, including overhead where required. Define the ratio direction before computing it. Repeated-run encoding saves space on long runs but can expand short runs. Reduced precision or representative bins usually discard information. No compression scheme shortens every possible input while remaining lossless.

Worked example W40. An original 1000-byte file is encoded as a 600-byte payload plus a 100-byte header. Encoded size is 700 bytes. Define compression ratio as original/encoded: $1000/700=10/7$. Saving is $(1000-700)/1000=30\%$. For run-length encoding of AAAABBCD using 8 bits for a symbol and 8 bits for its count, there are four runs, costing $4\times16=64$ bits. The original eight characters also cost 64 bits; an 8-bit header makes the encoding 72 bits, so it expands. Lossless encoding must permit exact reconstruction; replacing a bin's values by their mean usually does not. No lossless scheme shortens every possible input: four two-bit strings cannot all be mapped distinctly into the three shorter strings empty,0,1. Overhead and the data's pattern both matter.

Check before moving on. Include all overhead. State the ratio's direction and whether exact reconstruction is possible.

Previous-year questions, worked

Read the original question and options in the linked official paper. These are original explanations, not replacement question statements. PDF page numbers count from the first page of the downloaded file.

DA-2024-Q27 · Z-score with consistent units

Official question, PDF page 17. Published key: **B**.

Use the mean 96,000 and standard deviation 21,000, not the minimum and maximum. For 106,000, $z=(106000-96000)/21000=10/21\approx0.47619$, matching B. The result is dimensionless because the numerator and denominator use the same units. Round at the end.

Chapter practice

Original questions. Write the answer and the intermediate reasoning. Each lettered part belongs to the same problem. The separate solutions guide contains hints, steps and a final check.

P37 · Classification and calculation

(a) Classify department code, Low/Medium/High rating, Celsius temperature and mass in kg as nominal/ordinal/interval/ratio. (b) For x=60,min=20,max=80,mean=50,SD=10, compute min-max to [0,1], min-max to [-1,1] and z-score. For decimal scaling use max absolute value 80 and the smallest j with max|x|/10^j<1. (c) What breaks if all observations equal 7?

P38 · Bins and smoothing

Data are 0,1,2,3,8,9. (a) Use intervals [0,3),[3,6),[6,9] and replace each observation by its bin mean. Give memberships and the smoothed sequence. (b) Use equal-frequency bins of two sorted observations each and repeat. Explain how assigning bin labels instead would change the output type.

P39 · Sampling

Strata A and B have population sizes 90 and 10. Five sampled observations from each stratum have means 10 and 30. (a) Estimate the population mean with appropriate weights and compare the unweighted sample mean. (b) Count two ordered draws from five IDs with and without replacement. If a design selects villages and includes all their households, what is the first-stage sampling unit?

P40 · Compression

(a) Original size is 1000 bytes. Encoded payload is 600 bytes and header 100 bytes. Give total encoded size, original/encoded ratio and percentage saved. (b) Encode AAAABBCD as (symbol,count) runs with 8 bits per symbol, 8 per count and an 8-bit header. Compare encoded and original sizes. Why can no lossless code shorten every two-bit input?

Before the exam

Match the numerical operation to what it preserves or loses. Use units consistently. Keep sampling weights and compression overhead visible. These are data transformations, not database schema-normal-form problems.